



Oral Surgery

Lec.14

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Pre-extraction evaluation

- PEE is very valuable and necessary for successful extraction practice.
- Hurry and inadequate PEE of the case may lead to many **embarrassing difficulties intra-operatively** for the operator, in addition to the **postoperative** problems to the patient.

Pre-extraction evaluation

- Good P.E.E resulted from data collected from history, clinical examination, radiographs and laboratory aids in addition to that P.E.E need good knowledge and experience in the basic sciences e.g. **anatomy, physiology, pathology.**

In general, P.E.E may help you in: (Objective)

- a) Determine the method of extraction and type of anesthesia.
- b) Reduce the time spend for extraction
- c) Reduce the risks of intra and post operative crisis and complications.

P.E.E include:

- 1-Clinical preoperative evaluation.
- 2- Radiological evaluation.

Clinical P.E.E includes:-

A) General evaluation.

1. General impression of the patient.
2. History of general diseases, Nervousness and orientation
3. General oral hygiene.
4. Gingival inflammation, calculus, neglected mouth.

B) local evaluation.

1. Clinical examination to the accused tooth.
2. Adjoining structures.
3. Access to the tooth.

This includes the mouth opening, location of the tooth (eg, buccally malposed) may present difficulty in positioning the dental forceps for extraction, so the decision may surgical extraction.

4. Tooth mobility: The mobility of the tooth to be extracted should be assessed preoperatively, greater than normal mobility is frequently seen with **sever periodontal disease** but sometimes it maybe because of the **underlying pathology** like neoplasm

5- Condition of the tooth: - e.g.

- a)** Carious destruction.
- b)** The presence of large restoration.
- c)** Presence or absence of the adjoining teeth.
- d)** Non-vital tooth.
- e)** State of the supporting tissue.
- f)** Shape, position, long axis and size of the crown.
- g)** Attrition.

Radiographical evaluation

- Preoperative clinical assessment may be supplemented some times by preoperative radiographs, and the positive indications for preoperative radiograph are:
 1. History of difficult or unsuccessful extraction.
 2. Crown with extensive caries, large restorations
 3. Non-vital tooth when diagnosis is not certain.
 4. Malposed tooth.

- 5) A tooth which is abnormally resistant to forceps.
- 6) If after clinical examination you decide to remove the tooth surgically.
- 7) Any tooth which is in close relation to important or vital structures like **neurovascular canal**, **maxillary sinus**, mental nerve, nasal cavity.
- 8) If a tooth is **partially erupted** or **completely unerupted** or retained root.



- 9) An **isolated maxillary molar** especially if it is unopposed and over erupted. The bony support of such a tooth is often weakened by the presence of maxillary sinus and this may predispose to certain of **oro-antral communication** or fracture of the maxillary tuberosity.

The maxillary sinus can normally be seen above the level of the premolar and molar teeth in the [upper jaw](#). This [dental x-ray film](#) shows how, in the absence of the second premolar and first molar, the sinus became pneumatized and expanded towards the crest of the alveolar process (location at which the bone meets the gum tissue).



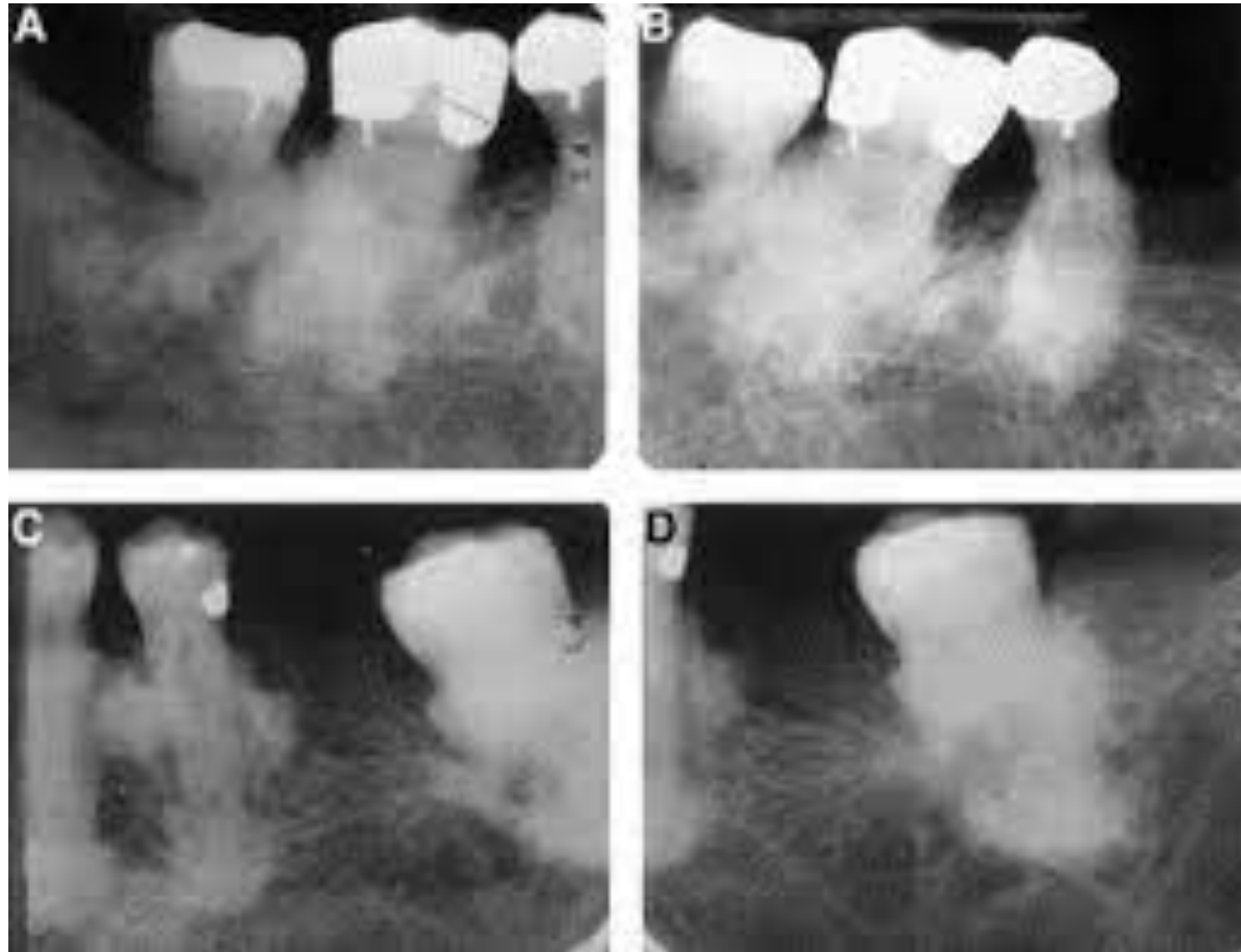
- 10) whenever, underlying bony pathology is suspected eg, **cystic lesion, tumors.**
- 11) Attrited teeth in elderly patient (maybe associated with hypercementosis).





- 12) Any systemic condition which may predispose to dental or alveolar abnormality like:
 - **a)** Osteitis deformans (Paget's disease), in which the roots are hypercementosed and ankylosed leading to difficult extraction and infection of the socket.
 - **b)** Cleido-cranial dysostosis, for pseudo-anodontia (multiple impactions, hooked roots occur, supernumerary teeth).

Paget's disease





- **c)** Patient who have received therapeutic irradiation to the jaw which may have to predispose to osteoradionecrosis.
- **d)** Osteopetrosis (marble bone disease), which cause extraction very difficult and predispose to chronic osteomyelitis.

A good radiograph and careful interpretation may give or aid the operator to many factors that may cause difficult extraction, e.g.:-

- 1- Abnormal number and shape of roots.
- 2- An unfavourable root pattern.
- 3- Caries extending to the root mass.
- 4- Fracture or resorption of the root.
- 5- Hypercementosis of roots.
- 6- Ankylosis (there is no space in periodontal ligament), and sclerosis of the bone.
- 7- Gemination (the development of two teeth from one bud).
- 8- Impaction.
- 9- Bony sclerosis and pathological lesions.